

THE LEVELING LINE

THE LEVELING LINE

A3 BOOKLET

A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.

Depart from a known point, carry the height station by station, run the circuit, return — and the difference is the closure error.

Within tolerance, every station on the line is accepted;

over tolerance, the whole run is void and re-measured, and no single station may be patched.



That hundred-year-old rule answers this belt's hardest question:

on what basis should a city trust its artificial intelligence — not on how well it performed once, but on whether it measures back.

Two readings of the same point, and the trust metric for this corridor.

CONTENTS

- | | | | |
|---------------|--|---------------|---|
| FIG.00 | The Leveling Line: a city that publishes its own error | FIG.06 | Recomputed metrics and the field census |
| FIG.01 | Overall concept and site cross-check | FIG.07 | Identity: mark, construction and applications |
| FIG.02 | Evidence chain and submission package: a leveling circuit not yet closed | FIG.08 | Innovation ecosystem and element mechanisms |
| FIG.03 | Three scope levels and network orders | FIG.09 | Landmarks, kit of parts, signage syntax and operating cycle |
| FIG.04 | Three key areas and benchmark layout | FIG.10 | The benchmark on the street, and how the kerb is shared |
| FIG.05 | Slow mobility, blue-green and closing routes | FIG.11 | Regional coordination interfaces: extending the levelling network |

Three AI pilgrimage landmarks

THREE LANDMARKS • spoken in engineering language; no spectacle

L1 The origin benchmark stone

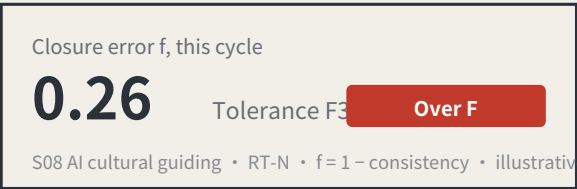
BM-0 • AI Origin Community



Every merged proposal's number is set into the paving — contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

L2 The closure stele

BM-1 • Zhongzhiyuan



A public reading wall, updated live, showing each scenario's closure error against its tolerance. Exceedances are marked in datum red. **The value of this landmark is that it is allowed to look bad.**

L3 The zeroing point

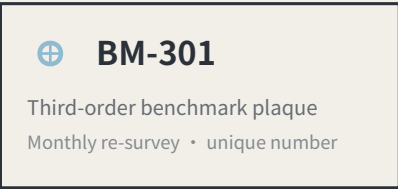
BM-2 • Dazhongsi



An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

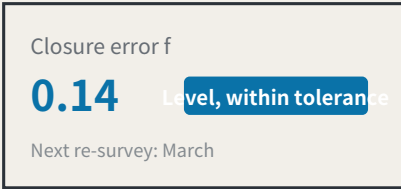
Kit of parts for public space: five standard components

KIT OF PARTS • common specifications and open drawings, so any new node joins in one language



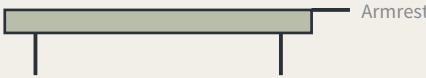
1 Stone and plaque

The number is the position, the order and the cycle Visible on site; no internet needed



2 Reading board

Visible on site; no internet needed



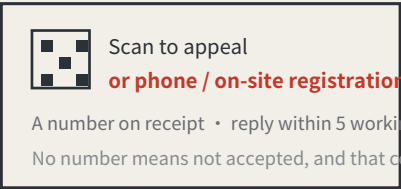
3 Dwellable seating

With armrests: older people rely on them to stand



4 Step-free guidance

Read by the users themselves



5 Complaint entry

A non-scan method is mandatory, or the right does not exist for P4

Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

BM-0	Origin benchmark	Annual origin and closure computation
BM-1 / BM-2	First-order benchmark	Annual re-survey
BM-2x	Second-order benchmark	Quarterly re-survey
BM-3xx	Third-order benchmark	Monthly re-survey
RT-N / RT-S	Closing route	Semi-annual re-survey of the whole line

Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

Monthly	Community day	Third-order BM-301/302/303	Led by P4 older residents, P5 wheelchair users, P7 frontline workers
Quarterly	Scenario open day	Second-order BM-21/22	Led by P2 developers and P3 students and faculty
Semi-annual	Route re-survey	Closing routes RT-N / RT-S	Led by professional bodies, with all four review categories present
Annual	Zeroing ceremony	3 The zeroing point	Readings zeroed, tolerance revisions read out, returned scenarios accounted for

Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate
International communication uses published readings as its material, never commitments.

The f=0.26 / f=0.14 on the reading boards are illustrative, not measured results. f is always a deviation: larger is worse, and $f \leq F$ is the passing test. Landmark siting requires heritage and engineering review; this proposal does not pre-empt it. The components are a directional proposal for a professional team to develop and are not construction drawings.

The chain of custody for innovation elements

ELEMENTS AND THEIR CHAIN OF CUSTODY

Every element answers the same three questions: which space carries it, which benchmark holds its reading, and what gets re-measured.

Element		Spatial carrier		Benchmark		Re-survey item
Land and space	→	Key-area renewal parcels	→	BM-1 / BM-0 / BM-2	→	Delivery rate of promised space
Compute	→	Consolidated compute nodes	→	BM-1	→	Share of public compute open to application
Data	→	Data-factor circulation venue	→	BM-2	→	Authorisation-chain completeness
Scenarios	→	Real-use venues in the two wings	→	BM-21 / BM-22	→	Enforceability of scenario exit
Funding	→	Zhongguancun technology-services wing	→	BM-21	→	Deliberately blank — see the note below
Talent	→	Talent district and third places	→	BM-0	→	Re-survey of service satisfaction

The funding row has no re-survey item. That is a judgement, not an omission.

Investment arrangements are decisions for market actors; an urban design proposal must not turn them into governance metrics, still less into fiscal commitments.

Writing what is not yours to govern into a metric system is its own kind of overreach.

Six global cases, one question

SIX CASES, ONE QUESTION: WHAT MAKES ITS TRUST RE-MEASURABLE?

Numbering	Case	Trust mechanism	Re-measurable	Transferable point
C1	Algorithm registers	Public register of purpose, data, owner and appeal route	High	The information base for a benchmark plaque
C2	Risk-tiered AI legislation	Duties differentiated by risk class	Medium	Supports the tiered setting of tolerance F
C3	Standardised testing frameworks	Comparable reports from a common toolkit	High	Gives re-survey its technical form
C4	Algorithmic impact assessment practice	Mandatory ex-ante and ex-post review for high-risk uses	led-high	Supports the strictest tolerance for health scenarios
C5	Civic data-stewardship practice	Data use decided by a citizen agenda	Medium	Supports public re-survey rights at third-order points
C6	Open-source reproducibility norms	Conclusions must ship a re-runnable artifact	High	This package ships a runnable verifier

All six point at the same gap: they register and they assess, but none institutionalises returning to the origin and computing.

A register says what a system declared; an assessment says what experts think. Neither answers how much the conclusion differs when the same public question passes through different nodes at different times.

Case material is drawn from public institutional documents and public reporting. Mechanisms only are cited: no text or images copied, no marks used, no non-public data, and no fabricated company lists, investment figures or output values. Element-to-space correspondence is in geometry/land_use.geojson and public_space.geojson.

Method: trust comes from closure, not from one accurate reading and add one check to the gates.

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so $f \leq F$ is always the passing test.

Act One: measure this open call first

228	Merged proposals (git tree, authoritative)
30.3%	Machine-readable disclosure field empty
84 → 8	Distinct strings written → model families

The “machine-readable” disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail,

Act Two: turn the same instrument on the site

412.5 m	Provisional design area to surveyed park
100%	Agreement with the research area

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km² level agrees completely; the discrepancy is confined to the 11.4 km² level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

8 / 8	Cross-jurisdiction points on this belt
-------	--

Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

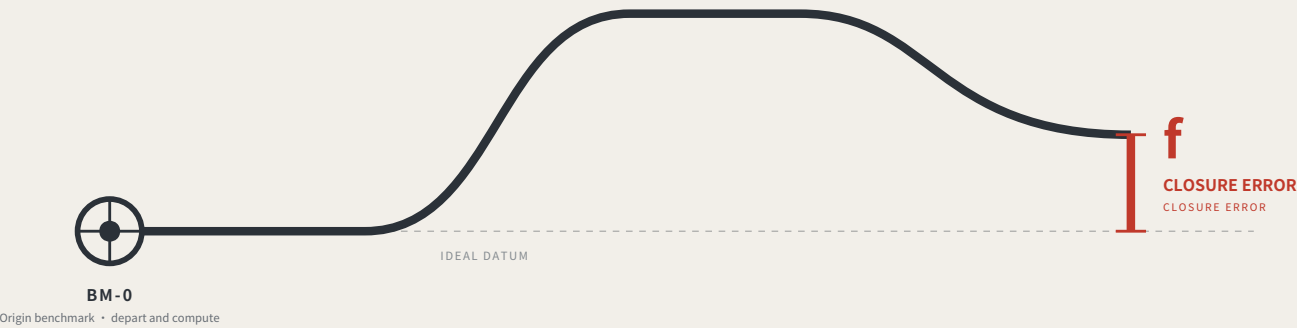
Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.

THE LEVELING LINE

THE LEVELING LINE

A city that publishes its own error.

A city that publishes its own error.



Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

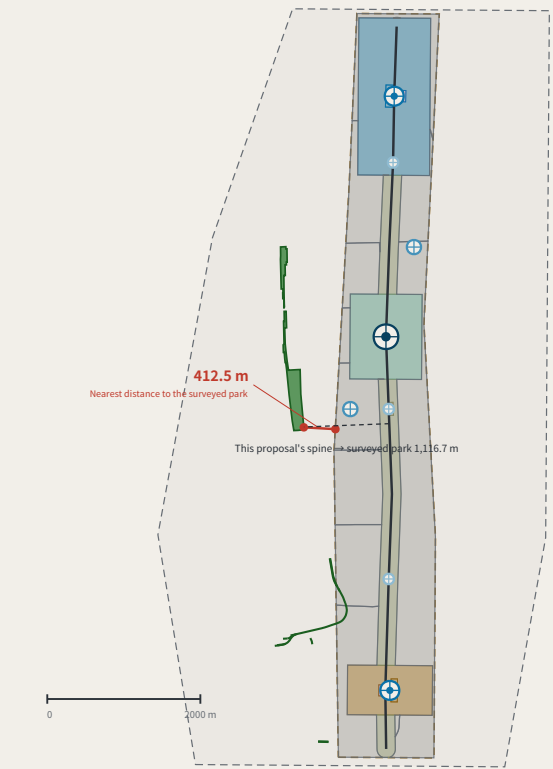
This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

jiangmuran • Open call for the Centennial Jing-Zhang AI Innovation Belt • Concept advice on provisional boundaries; not a substitute for statutory planning

FIG.00

OVERALL CONCEPT AND SITE CROSS-CHECK

Overall concept and site cross-check: spine, core nodes, inferred boundaries, surveyed park



Legend and how to read it

LEGEND • surveyed data and inferred boundaries are shown separately

Surveyed (independently checkable)

- Jing-Zhang Railway Heritage Park (OSM survey, 17.49 ha)
- Disused railway (14 segments, 3,028 m)

Inferred (not usable as a redline)

- Coordinated research area 43.6 km²
- Overall design area 11.4 km²

This proposal's design (generated against inferred boundaries)

Blank-block edges follow the existing arterials; they are not a parcel subdivision or an ownership boundary

- Cultural use (0803)
- AI R&D and research (0802)
- Community services and support (0702)
- Industry and commercial services (05)
- Park and green space (1401)
- Urban road land (1207)
- Left blank by this proposal (code 16)
- Spine (design centreline)
- Tiered benchmarks

The line to read on this sheet is the red one.

Research area and surveyed park agree 100%, so the announced bounds do not conflict with actual geography; the discrepancy is confined to the overall design area. This proposal's spine is displaced with it. Marked here rather than hidden.

Two independent routes to the same object

TWO INDEPENDENT ROUTES TO THE SAME OBJECT

17.49 ha

Surveyed park area (OSM)

0%

Intersection with the provisional design area

412.5 m

Nearest distance between them

100%

Agreement with the research area

The same instrument also measured this proposal:

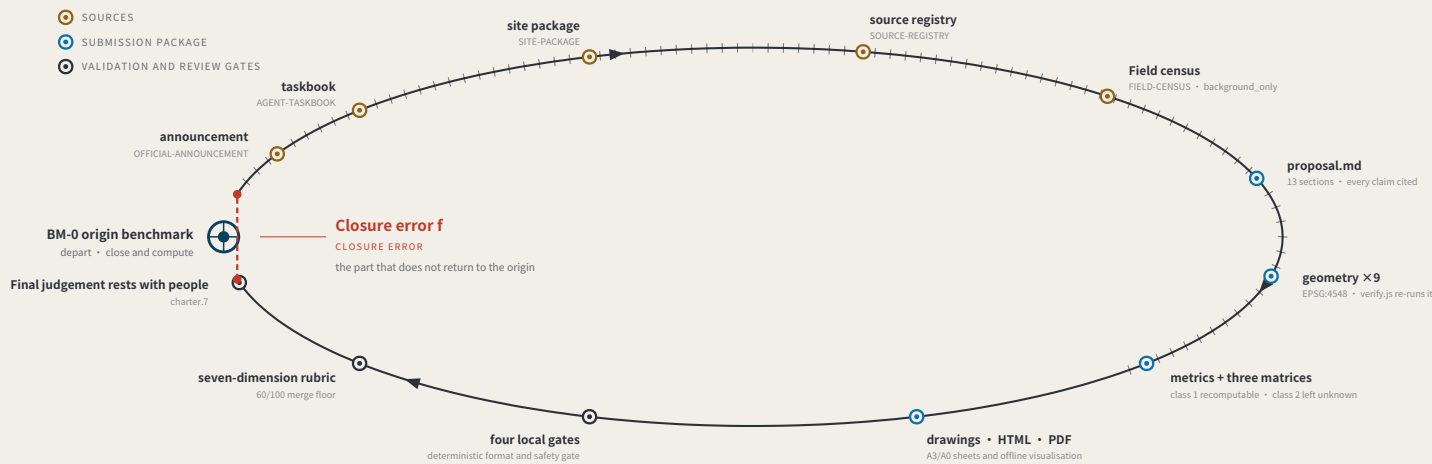
the submitted spine lies 1116.7 m from the surveyed park, because it was generated against the provisional boundary.

A recomputability standard invoked only when it flatters the author is not a standard.

Surveyed data © OpenStreetMap contributors • ODbL 1.0 • Overpass API 2026-08-08. Script and conventions in visual/assets/osm_reference.json; re-runnable. OSM is crowd-sourced and the park polygon may cover only the mapped section; the provisional boundary is itself inferred. This sheet reports the difference and does not adjudicate which is right.
THE LEVELING LINE • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning
Recomputed in EPSG:4548 • exchanged in EPSG:4326

EVIDENCE CHAIN AS AN UNCLOSED LEVELING CIRCUIT

Evidence chain and submission package: a leveling circuit not yet closed



Act One: measuring this open call itself

MEASURING THE OPEN CALL ITSELF • as of 2026-08-09 • 373 merged • 373/373 fetched • PR numbers past 1000

373

Merged proposals (git tree, authoritative)

The gallery index lists 309 at the same moment — 64 behind

28.4%

Machine-readable disclosure field empty

106 of 373 still hold the scaffold placeholder or are empty

117 → 8

Distinct strings written → actual model families

The GPT/Codex family alone is written 55 ways across 168 proposals

The "machine-readable" disclosure field is not machine-readable.

charter.6 requires disclosing the generation method and charter.5 requires it structured and agent-readable, yet none of the four gates checks this field. Nobody can aggregate "which models produced this call" from it.

The fix is light: split "model" into an enumerated family plus free-text detail, and add one enumeration check.

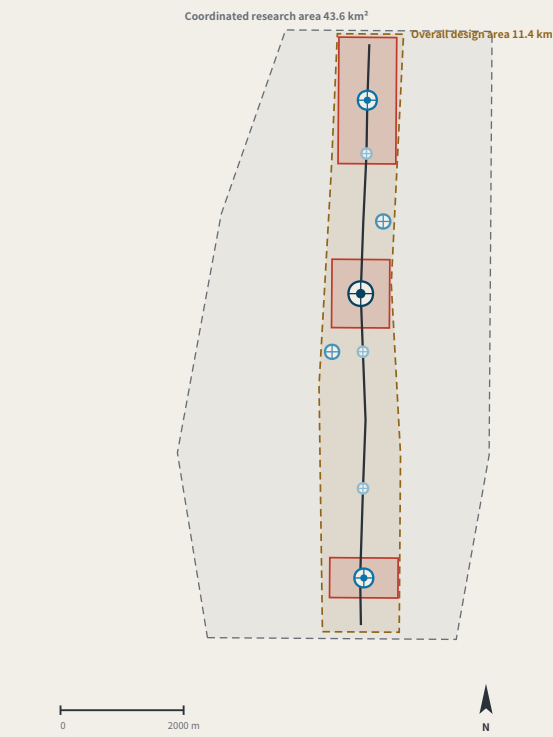
This proposal is the instrument for the missing last step: compute the closure error, and give it consequences.

Registers answer "what did the system declare" and assessments answer "what do experts think"; closure error answers "how much does the conclusion differ across nodes and across time for the same public question".

Sources: visual/assets/census.json, field_map.json and the first-round (184-file) snapshot agent_declarations.json, all shipped with the package. The authoritative source is the GitHub git tree, not the gallery index, which lags. Generation scripts are in the accompanying issue and are re-runnable. The corpus grows daily; recompute before citing.
THE LEVELING LINE • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning
Recomputed in EPSG:4548 • exchanged in EPSG:4326

THREE SCOPE LEVELS AND NETWORK TIERS

Three scope levels and network orders



THREE SCOPE LEVELS MAPPED ONTO SURVEY ORDERS

SCOPE LEVELS AND NETWORK TIERS

Announcement level	Scale	Network role	Re-survey cycle
Coordinated research area	43.6 km ²	Whole-network control	Annual
Overall design area	11.4 km ²	First-order route: spine plus two closing routes	Semi-annual
Key areas	368.4 ha	Origin BM-0 with first-order BM-1 / BM-2	Quarterly

The three levels are not three unrelated drawing sets

The research area decides what to measure.

The design area decides which route to measure along.

The key areas decide where to set the stones.

Benchmark orders

BENCHMARK ORDERS

- Origin benchmark Annual origin and closure check
- First-order benchmark Annual re-survey
- Second-order benchmark Quarterly re-survey
- Third-order benchmark Monthly re-survey

Land-use partition composition

LAND-USE PARTITION

22 features • 7 classes • no overlap, no gap (verified point by point)

Left blank by this proposal (code 16)	55.4% • 11 blocks
AI R&D and research (0802)	16.9% • 2 blocks
Park and green space (1401)	12.2% • 1 blocks
Community services and support (0702)	8.6% • 1 blocks
Industry and commercial services (05)	6.3% • 2 blocks
Cultural use (0803)	0.6% • 4 blocks
Urban road land (1207)	0.1% • 1 blocks

Reserved land at 55.4% is a judgement, not a gap: tenure, plan and retain/demolish status are unpublished; this proposal does not subdivide for them.

Values this proposal deliberately does not give

Floor area ratio • building height • density • setbacks • road redlines

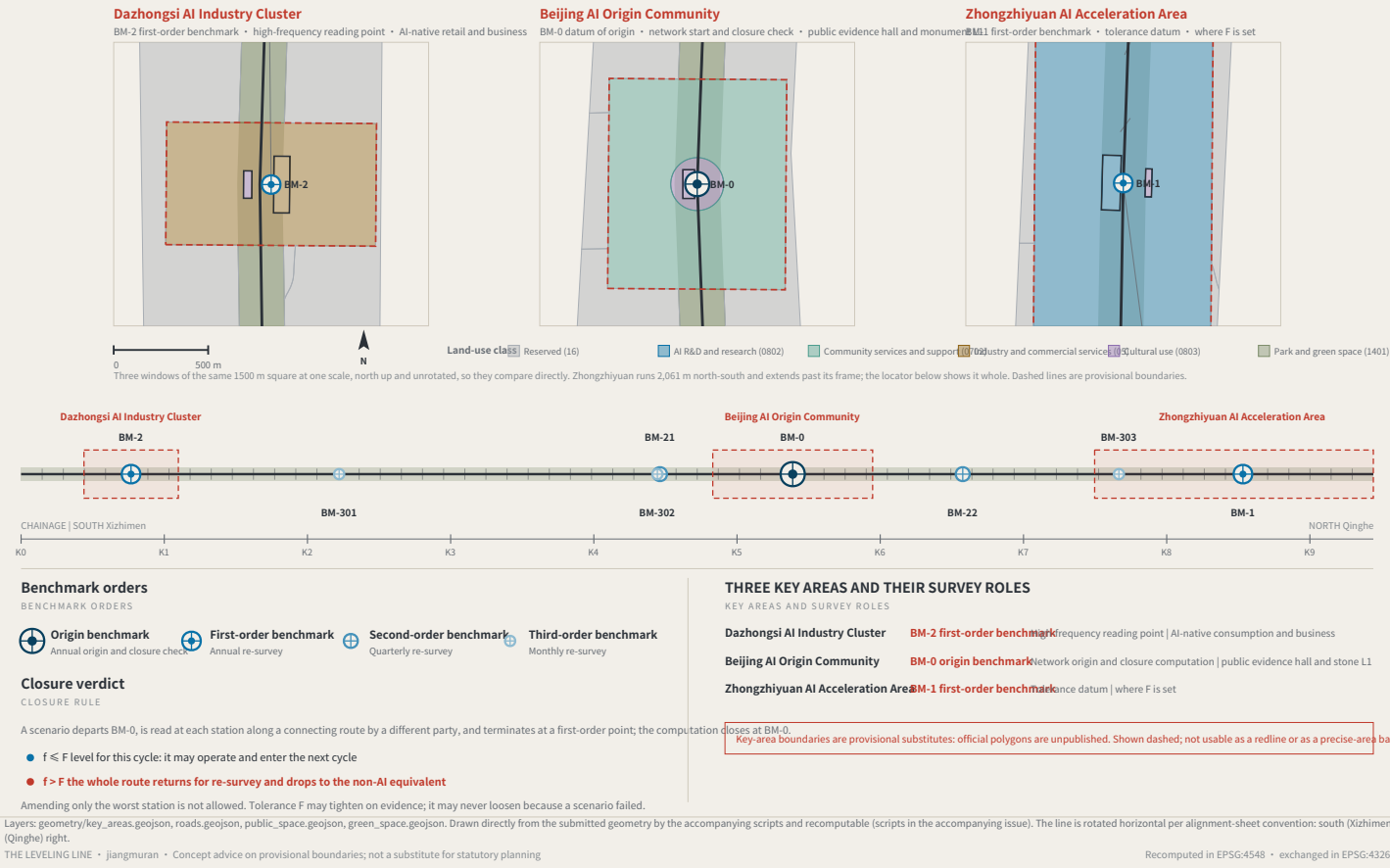
These are statutory regulatory-plan controls. Publishing a figure where the official data is absent would be manufacturing certainty; metrics.json keeps them unknown.

Layers: brief/site-package/geometry/provisional_boundaries.geojson, geometry/roads.geojson, public_space.geojson. Drawn by the accompanying scripts and recomputable (scripts in the accompanying issue). The three areas are taken from the announcement text; the geometry is a provisional substitute.
THE LEVELING LINE • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning
Recomputed in EPSG:4548 • exchanged in EPSG:4326

KEY AREAS AND BENCHMARK LAYOUT

FIG. 04

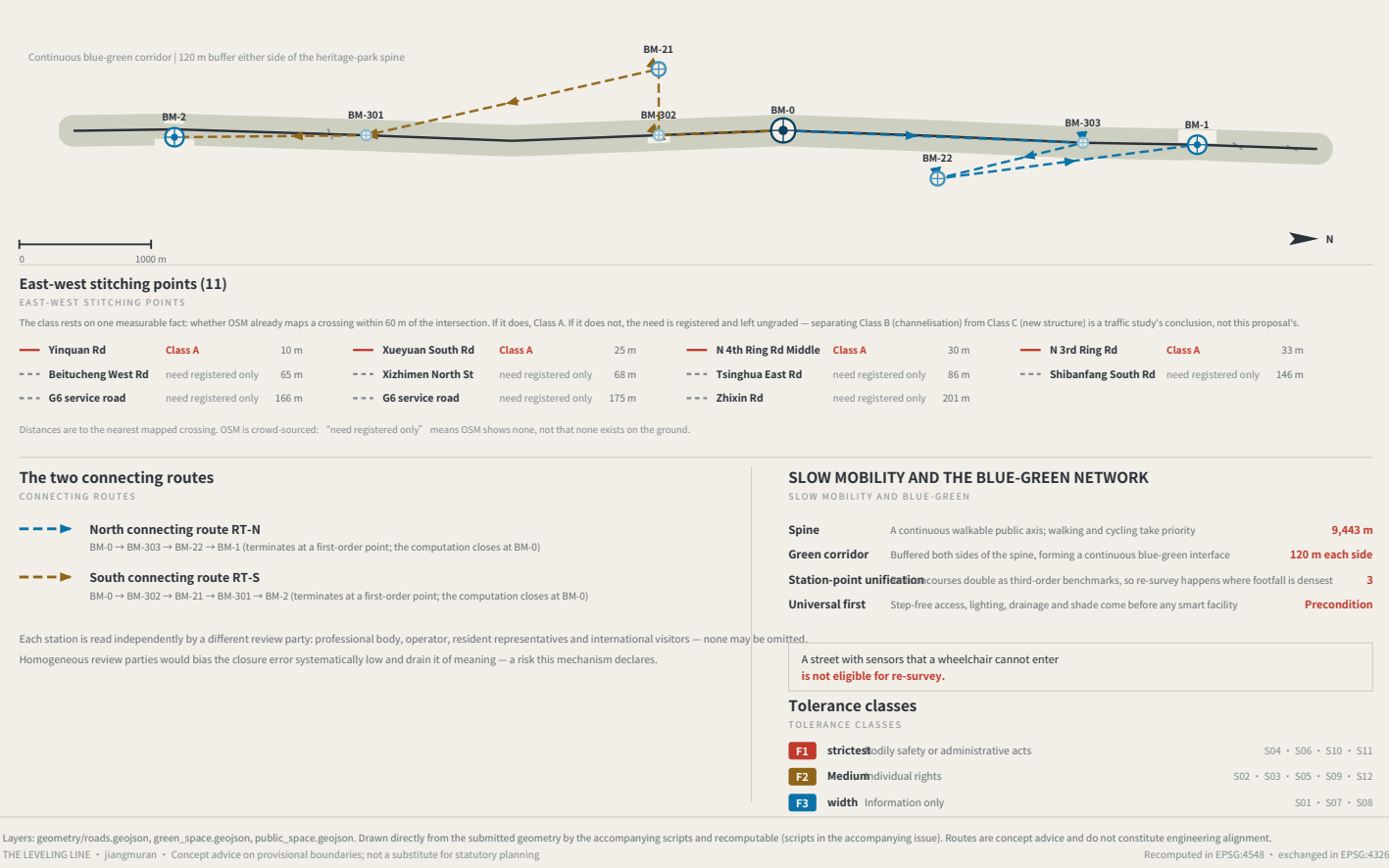
Three key areas and benchmark layout



SLOW MOBILITY, BLUE-GREEN AND CONNECTING ROUTES

FIG. 05

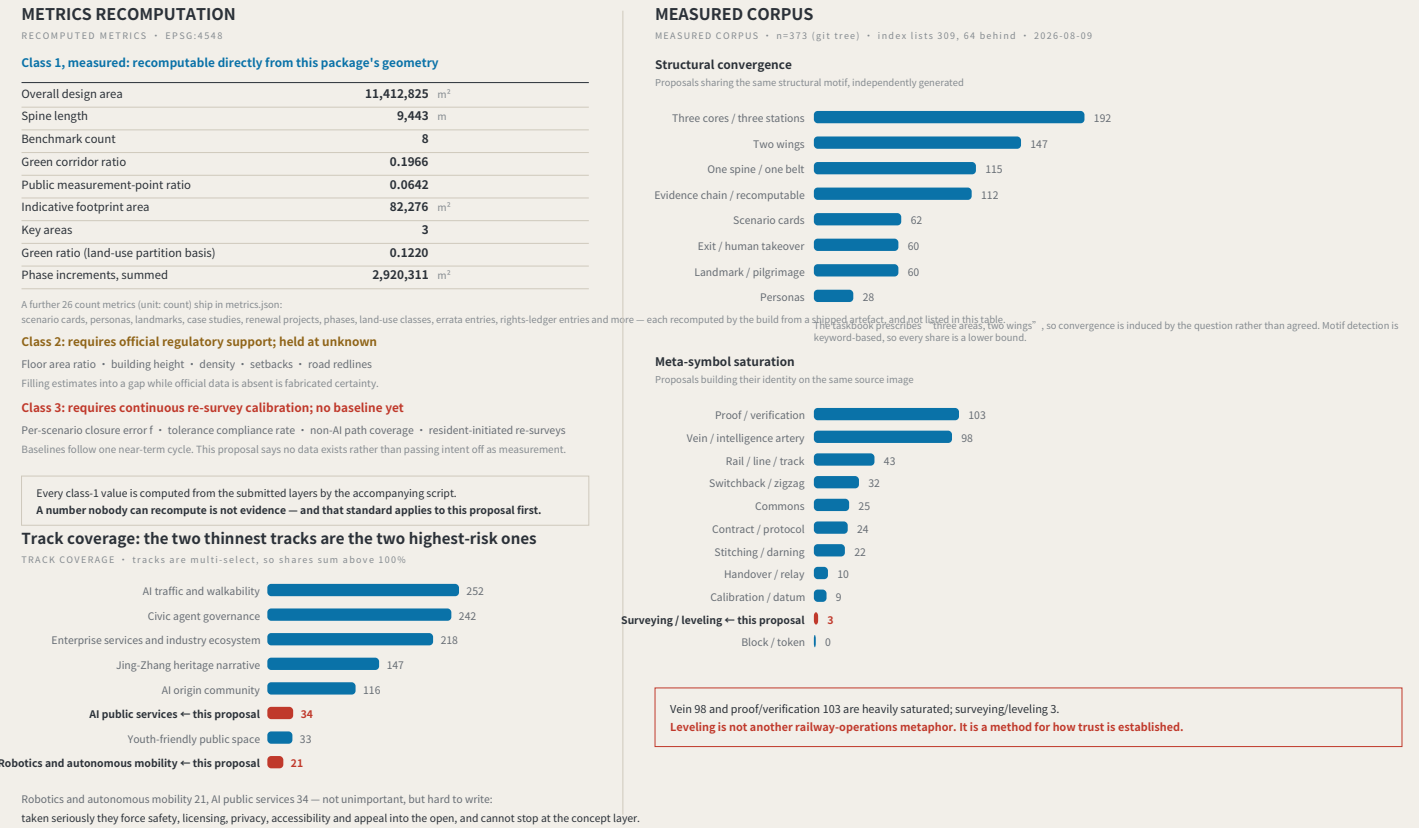
Slow mobility, blue-green and closing routes



RECOMPUTED METRICS AND MEASURED EVIDENCE

FIG. 06

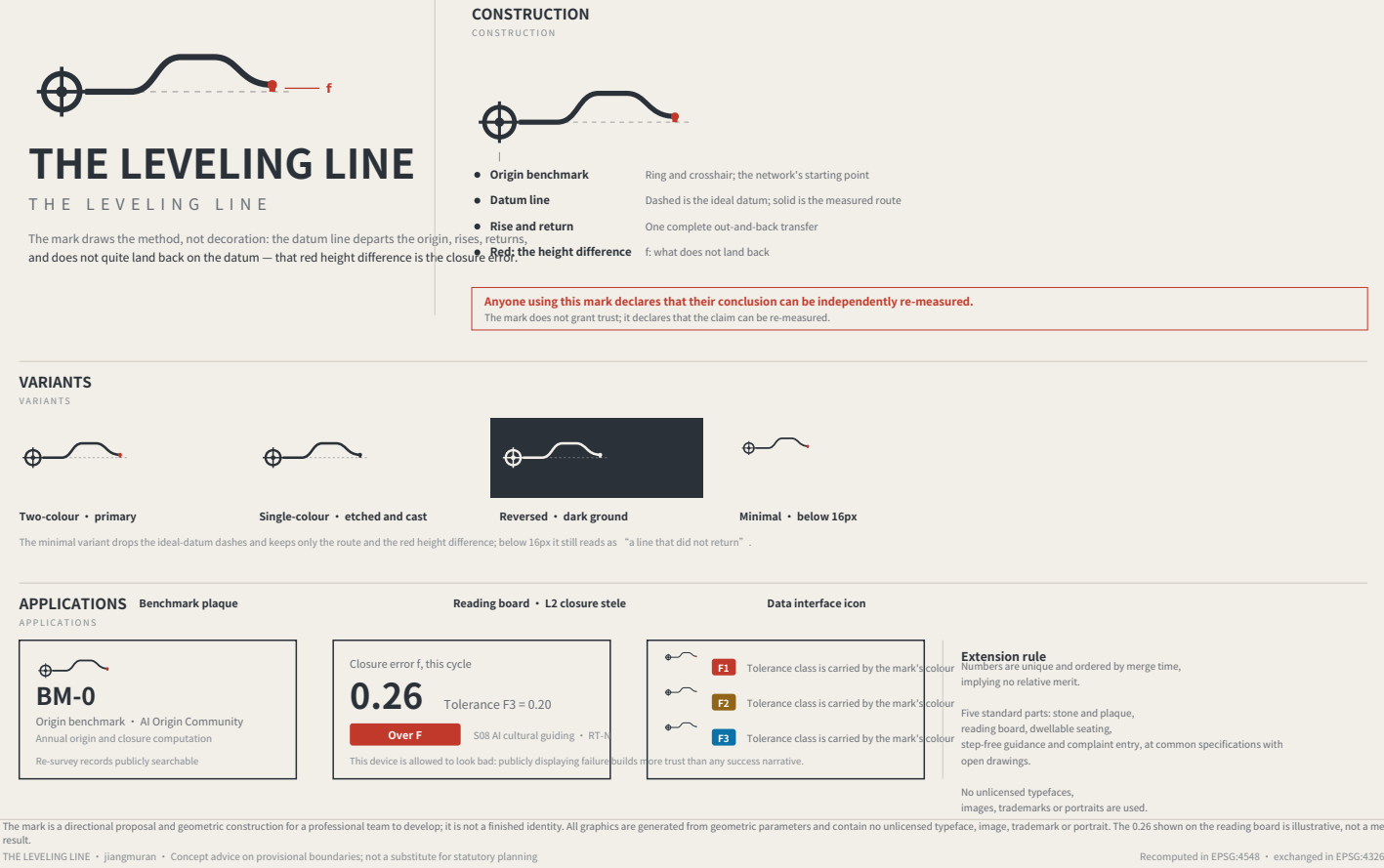
Recomputed metrics and closure-error evidence



IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

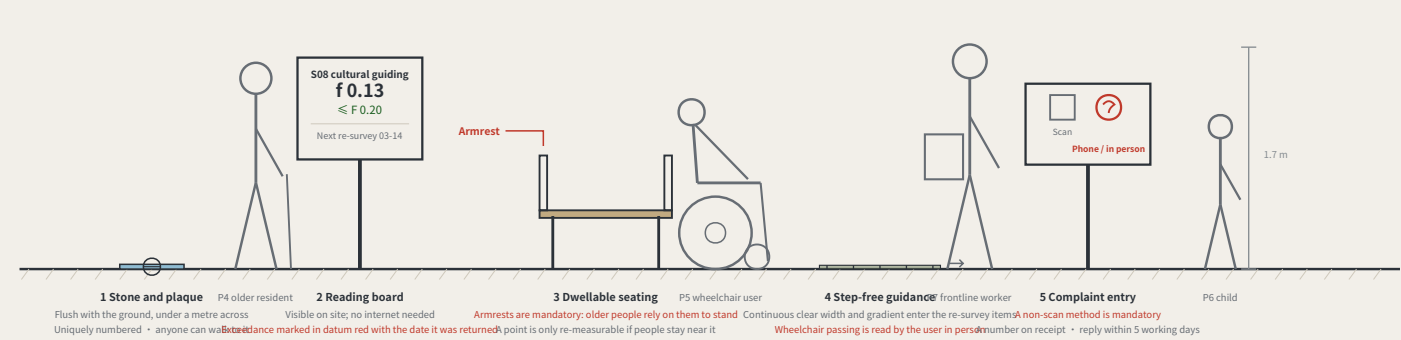
FIG. 07

Identity: mark, construction and applications



What a benchmark looks like at eye level

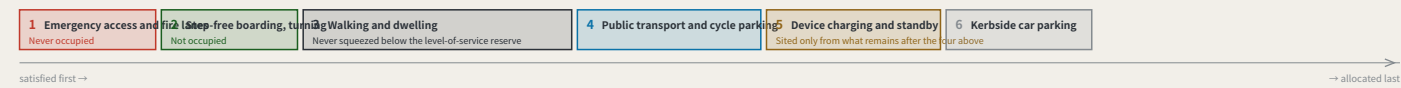
A BENCHMARK AT EYE LEVEL • third-order BM-301 • elevation 1 m = 138 units



Two of the five standard components are hard constraints: seating must have armrests, and the complaint entry must offer a non-scan method. A rule you cannot see in the drawing is a rule the drawing does not contain.

Kerb allocation in priority order

KERB ALLOCATION IN PRIORITY ORDER • sequence and precedence only, no widths



The order is itself a position: device charging sits behind pedestrians and accessibility, which means introducing devices must not be paid for with degraded walking conditions. Winter snow storage must be reserved at this stage alongside the device and pedestrian lanes — piled on the device lane the devices stop; piled on the pedestrian lane people are pushed toward the carriageway, which is worse.

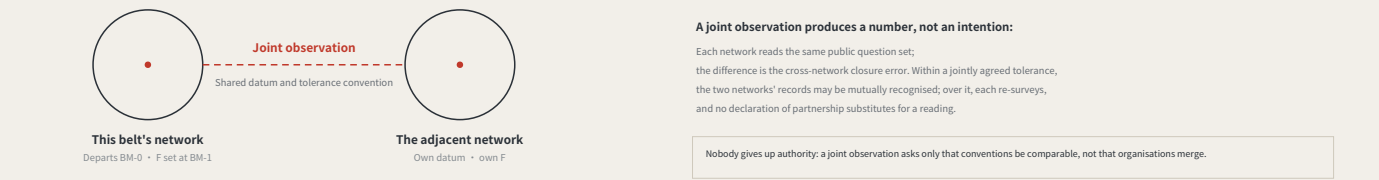
The elevation is drawn at 1 m = 138 units. The figures are a scale reference and a statement about who is present, not a rendering. The section expresses sequence and precedence only and carries no widths — capacity must be computed on site from measured clear width, and a figure not derived from measurement is fabricated certainty. Component specifications are in geometry/public_space.geojson and the scenario cards; positions await official boundaries and title verification.

THE LEVELING LINE • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning

Recomputed in EPSG:4548 • exchanged in EPSG:4326

An executable form of coordination: how two levelling networks join

AN EXECUTABLE FORM OF COORDINATION: HOW TWO LEVELLING NETWORKS JOIN



What each of the five partners exchanges, and what it does not

Partner	Stage	Exchanges	Explicitly does not exchange	Precondition for the interface
BDA (Yizhuang)	Volume production and high-level AV road testing	Mutual recognition of closure records across complementary speed domains	Not one admission: an F1 pass here admits a device to pedestrian density, not to a carriageway, and the reverse also holds	Both parties confirm the speed-domain split and test protocols
Future Science City	Corporate research institutes and engineering	Scenario demand and real user density	No claim on IP; no scheduling on anyone's behalf	Wing benchmarks built and a first baseline taken
Huairou Science City	Large facilities and basic research	The measurement method itself: how f is defined, cycle length, the statistics behind revising F	No scenario data is exchanged; the coordination is methodological, not applied	The closure convention published and open to methodological review
Beiwei Community	The innovation community the taskbook names	Reciprocal joint points; exchanged review parties	This proposal does not know its positioning and does not set its benchmark order	Each party names at least one publicly reachable joint point
Beijing-Tianjin-Hebei	Cross-provincial coordination	Cross-provincial recognition of the tolerance convention	No direct recognition of records; without a shared convention they are not comparable	Tolerance definition and revision procedure agreed across provinces

None of the five rows has been confirmed by the party named.

No internal plans are known, nobody is committed, no agreement is assumed; each row is an interface to evaluate independently.

Each partner's stage is described from publicly available general positioning and is unconfirmed by them. The speed-domain relationship in the first row must be settled against both parties' actual test protocols. This sheet proposes interfaces; it commits nobody and is no government determination.

THE LEVELING LINE • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning

Recomputed in EPSG:4548 • exchanged in EPSG:4326